

Vireo firmware v1.2 sleep-dim flicker

Field	Value
Published	2026-02-24
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Incident window	2025-12-09 to 2026-02-17
Status	closed

This is a blameless review. Names appear as actors in the timeline; causes are stated as system and process failures.

Summary

Firmware v1.2 shipped over the air on 2025-12-09 and introduced a visible 0.5 Hz flicker on Vireo smart desk lamps (LS-LMP-001) built with Meridian PCB rev C, whenever sleep-mode dimming brought brightness below 20 percent. Twelve customers filed tickets between 2025-12-11 and 2026-01-14. Support agents shipped a workaround within two weeks, a staged rollback to v1.1 contained most of the exposure in January, and hotfix firmware v1.3 closed the defect on 2026-02-17. One B2B account, Cobalt Dental Group, was affected on two units and is now fully resolved.

Impact

- 12 tickets against 618 exposed units, a 1.9 percent ticket rate.
- 592 of 618 units restored by the staged rollback (09-13 Jan); the remaining 26 recovered afterward through direct support contact or the v1.3 push.
- Cost: 4,800.00 under SOW Amendment No. 2 (INV-BGL-2026-006), plus the ongoing 2-year goodwill warranty exposure on all 618 vireo-v12-affected units.

Timeline

date	event
2025-12-09	firmware v1.2 released via the Bitgrove Labs (VEN-03) OTA service
2025-12-11	first flicker ticket, HD-2025-05402, filed by Jonah Beck
2025-12-18	pattern escalated internally: six identical tickets in seven days
2025-12-19	vendor bug report filed with Bitgrove Labs; same day, Cobalt Dental Group reports flicker on both of its add-on units
2025-12-22	customer troubleshooting guide published by Priya Raman
2026-01-06	follow-up bug report with debug log captures narrows the root cause
2026-01-09	staged rollback to v1.1 begins; 592 of 618 units restored by 13 Jan
2026-01-20	Mara Voss issues a 2-year goodwill warranty, scope vireo-v12-affected
2026-01-26	Ana Petrova executes SOW Amendment No. 2 with Bitgrove for the hotfix
2026-02-17	hotfix v1.3 released; v1.1 channel pin removed

Root cause

The v1.2 sleep-timer wake interrupt collides with the Meridian PCB rev C LED driver's PWM cycle whenever duty drops below 20 percent, producing the roughly 0.5 Hz flicker. Rev B boards use a different LED driver and are unaffected under identical firmware and dim-schedule conditions.

Contributing factors

- Bitgrove's test bench had no rev C hardware at the time v1.2 was built, so the release shipped untested against the board revision that ultimately triggered the defect.
- The affected population spans two receiving lots, LOT-2025-31 and LOT-2025-38, both shipped from October 2025 onward, which meant the earliest exposed customers were also our newest ones.

What went well

- The disable-sleep-dim workaround was confirmed on the fourth ticket (HD-2025-05461, 2025-12-15) and published as a customer-facing guide within a week, on 2025-12-22.
- The staged rollback restored 592 of 618 units within five days once Bitgrove delivered the signed v1.1 image.
- The goodwill warranty extension closed the coverage gap for every affected unit, including the two on the Cobalt Dental Group account, before the hotfix even shipped.

What went poorly

- The first six tickets were worked as individual complaints for roughly a week before the pattern was escalated internally on 2025-12-18, which delayed the first vendor bug report by several days.

- 26 units could not take the January rollback because they were offline, and needed either individual outreach or the v1.3 push in February to recover.
- The 2024 public help doc for the Vireo lamp was never updated to reflect the goodwill extension, so it still states the 1-year standard warranty with no mention of the vireo-v12-affected scope.

Follow-ups

action	owner	due	status
add rev C units to the Bitgrove test bench	VEN-03	2026-02-10	done
qualify Meridian (VEN-07) rev D	DO	2026-03-31	in progress
track warranty scope vireo-v12-affected in Helprise	PR	2026-01-26	done
update the public Vireo help doc for the warranty scope	PR	--	not started